DATABASE CONCEPTS ASSIGNMENTS

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COURSE NAME: Masters of Data Science

COURSE CODE: ISYS1055

Part D

Data Visualisation

TaskD.1

```
WITH VaccineData AS (
SELECT
c.Country_Name AS Country_Name,
vr.Date AS Observation_Date,
vr.Total_Vaccinations AS Total_Vaccinations
FROM
Vaccination_Record vr
JOIN
Country c ON vr.Country_ID = c.Country_ID
WHERE
vr.Date IN ('2022-02-22', '2021-04-22', '2023-11-11')
),
PivotedData AS (
SELECT
Country_Name,
MAX(CASE WHEN Observation_Date = '2022-02-22' THEN Observation_Date END) AS Date1,
MAX(CASE WHEN Observation_Date = '2022-02-22' THEN Total_Vaccinations END) AS VOD1,
MAX(CASE WHEN Observation_Date = '2021-04-22' THEN Observation_Date END) AS Date2,
MAX(CASE WHEN Observation_Date = '2021-04-22' THEN Total_Vaccinations END) AS VOD2,
MAX(CASE WHEN Observation_Date = '2023-11-11' THEN Observation_Date END) AS Date3,
MAX(CASE WHEN Observation_Date = '2023-11-11' THEN Total_Vaccinations END) AS VOD3
FROM
VaccineData
GROUP BY
Country_Name
)
SELECT
Date1 AS "Date 1 (OD1)",
Country_Name AS "Country Name (CN)",
VOD1 AS "Vaccine on OD1 (VOD1)",
Date2 AS "Date 2 (OD2)",
VOD2 AS "Vaccine on OD2 (VOD2)",
Date3 AS "Date 3 (OD3)",
VOD3 AS "Vaccine on OD3 (VOD3)",
CASE
WHEN VOD1 > 0 AND VOD2 > 0 AND VOD3 > 0 THEN
```

```
((VOD2 - VOD1) * 1.0 / VOD1) - ((VOD3 - VOD2) * 1.0 / VOD2)
ELSE
NULL
END AS "Percentage Change of Totals"
FROM
PivotedData
ORDER BY
Date1, Country_Name;
```

Vaccination.db - Query

new_table_name Create Query

table name...

Country
Data_Source
Manufacturer
US_State_Vaccination
Vaccination_AgeGroup
Vaccination_Record
Vaccine

Toggle helper tables

```
WITH VaccineData AS (
  SELECT
    c.Country_Name AS Country_Name,
    vr.Date AS Observation_Date,
    vr.Total_Vaccinations AS Total_Vaccinations
```

Use Shift + Up/Down to navigate recently-executed queries

Execute Export JSON Export CSV SQL Help Bookmarks +

Results (unable to determine) Permalink

Date 1 (OD1)	Country Name (CN)	Vaccine on OD1 (VOD1)	Date 2 (OD2)	Vaccine on OD2 (VOD2)	Date 3 (OD3)	Vaccine on OD3 (VOD3)	Percentage Change of Totals
2022-02-22	China	3134137	2021-04-22	1488292	2023-11-11	1841568	-0.7625050752905992
2022-02-22	India	90333264	2021-04-22	6355931	2023-11-11	428509210	-67.34843422417599
2022-02-22	Ireland	120000	2021-04-22	547901	2023-11-11	444154550	-806.0816110665826
2022-02-22	United States	653129954	2021-04-22	641295	2023-11-11	556584	-0.8669244661198021

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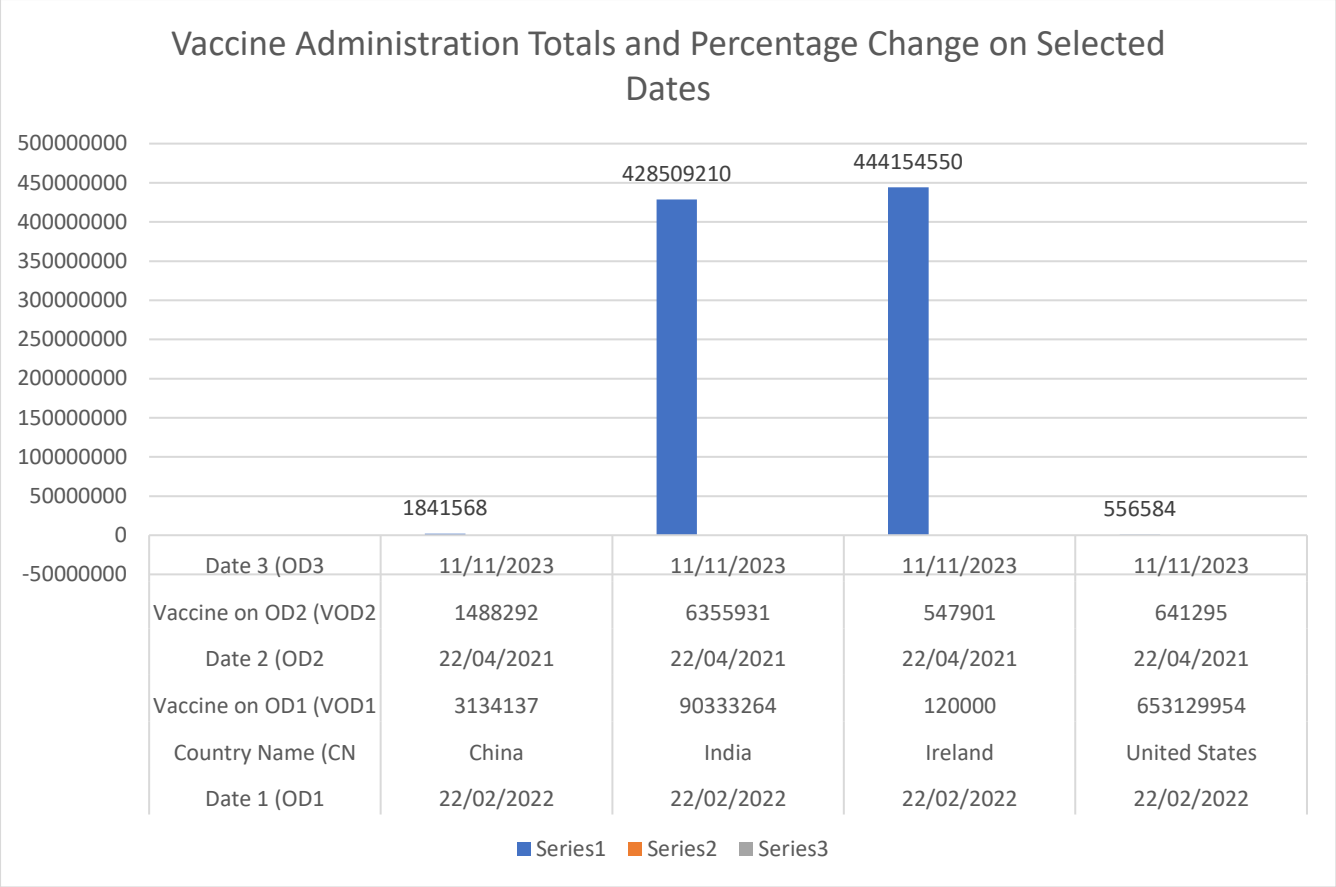

Task D.2.

```
WITH MonthlyData AS (
SELECT
c.Country_Name,
strftime('%Y', vr.Date) AS Year,
strftime('%m', vr.Date) AS Month,
SUM(vr.Total_Vaccinations) AS Total_Doses
FROM
Vaccination_Record vr
JOIN
Country c ON vr.Country_ID = c.Country_ID
GROUP BY
c.Country_Name, Year, Month
),

GrowthRates AS (
SELECT
Country_Name,
Year,
Month,
```

```
Total_Doses,
LAG(Total_Doses) OVER (PARTITION BY Country_Name ORDER BY Year, Month) AS Previous_Doses,
CASE
WHEN LAG(Total_Doses) OVER (PARTITION BY Country_Name ORDER BY Year, Month) > 0 THEN
(Total_Doses - LAG(Total_Doses) OVER (PARTITION BY Country_Name ORDER BY Year, Month)) *
100.0 / LAG(Total_Doses) OVER (PARTITION BY Country_Name ORDER BY Year, Month)
ELSE
NULL
END AS Growth_Rate
FROM
MonthlyData
),

GlobalAverageGrowth AS (
SELECT
Year,
Month,
AVG(Growth_Rate) AS Avg_Growth_Rate
FROM
GrowthRates
WHERE
Growth_Rate IS NOT NULL
GROUP BY
Year, Month
)

SELECT
gr.Country_Name,
gr.Month,
gr.Year,
gr.Growth_Rate AS Growth_Rate,
(gr.Growth_Rate - avg_gr.Avg_Growth_Rate) AS Difference_From_Global_Avg
FROM
GrowthRates gr
JOIN
GlobalAverageGrowth avg_gr
ON
gr.Year = avg_gr.Year AND gr.Month = avg_gr.Month
WHERE
gr.Growth_Rate > avg_gr.Avg_Growth_Rate
ORDER BY
gr.Country_Name, gr.Year, gr.Month;
```

Vaccination.db - Query

new_table_name Create Query

table name...

Country
Data_Source
Manufacturer
US_State_Vaccination
Vaccination_AgeGroup
Vaccination_Record
Vaccine

Toggle helper tables

```
WITH MonthlyData AS (
  SELECT
    c.Country_Name,
    strftime('%Y', vr.Date) AS Year,
    strftime('%m', vr.Date) AS Month,
```

Use Shift + Up/Down to navigate recently-executed queries

Execute Export JSON Export CSV SQL Help Bookmarks +

Results (6) Permalink

Country_Name	Month	Year	Growth_Rate	Difference_From_Global_Avg
China	06	2024	40.73631818102834	69.3825462290335
India	04	2021	535.6686753143364	471.44863827015433
Ireland	03	2021	3322.0267211436194	2081.0255562444995
Ireland	10	2022	7535358.906666666	5023604.501739129
United States	02	2022	101745.47735441568	76293.8838934112
United States	11	2023	190.5801163229516	199.3917669850312

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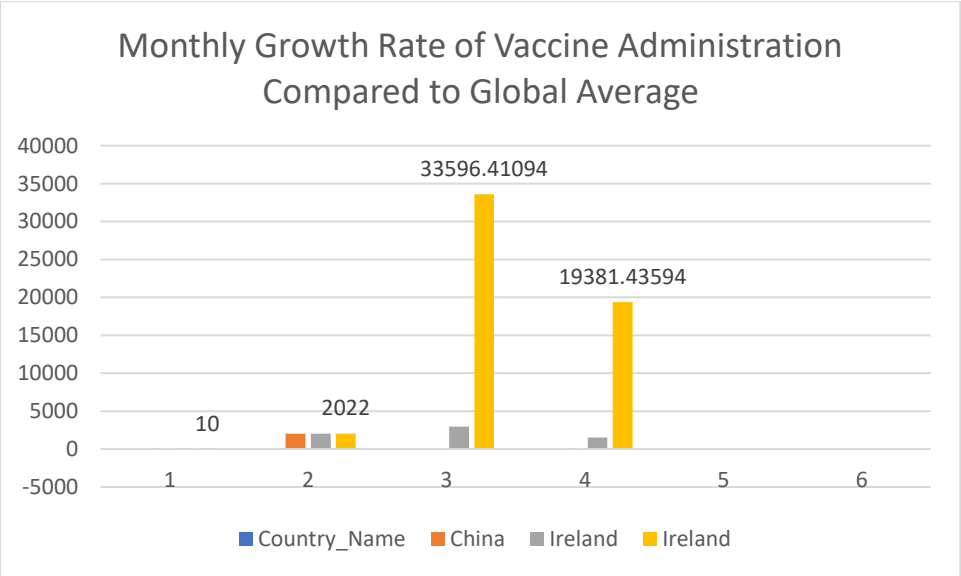


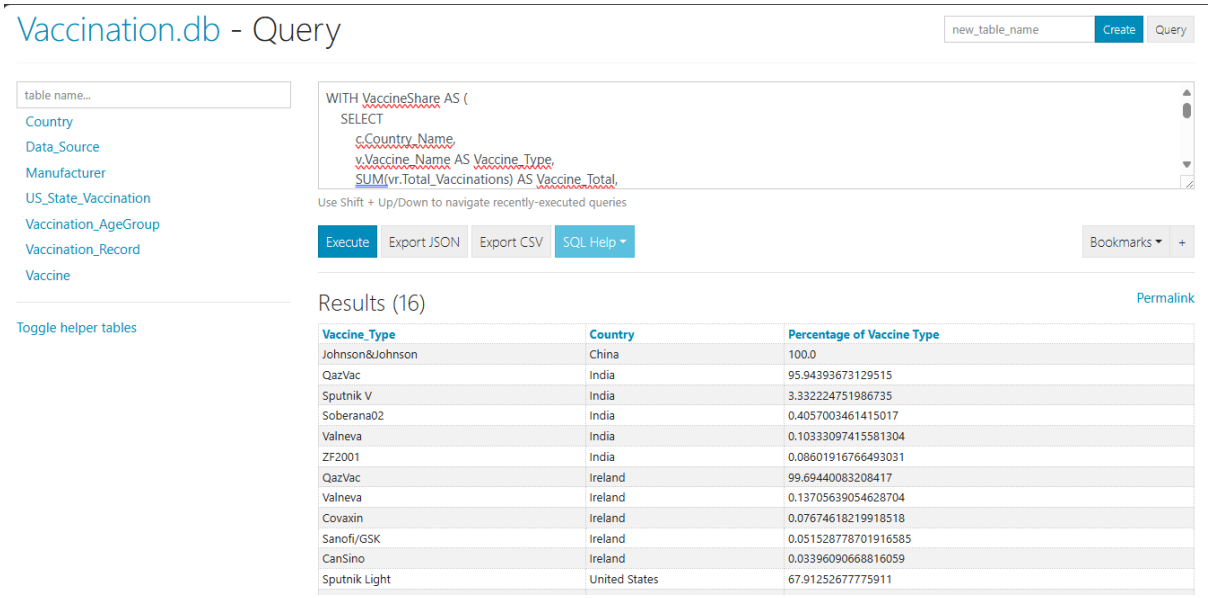
Task D.3.

```
WITH VaccineShare AS (
SELECT
c.Country_Name,
v.Vaccine_Name AS Vaccine_Type,
SUM(vr.Total_Vaccinations) AS Vaccine_Total,
SUM(SUM(vr.Total_Vaccinations)) OVER (PARTITION BY c.Country_Name) AS Country_Total
FROM
Vaccination_Record vr
JOIN
Country c ON vr.Country_ID = c.Country_ID
JOIN
Vaccine v ON vr.Vaccine_ID = v.Vaccine_ID
GROUP BY
c.Country_Name, v.Vaccine_Name
),
```

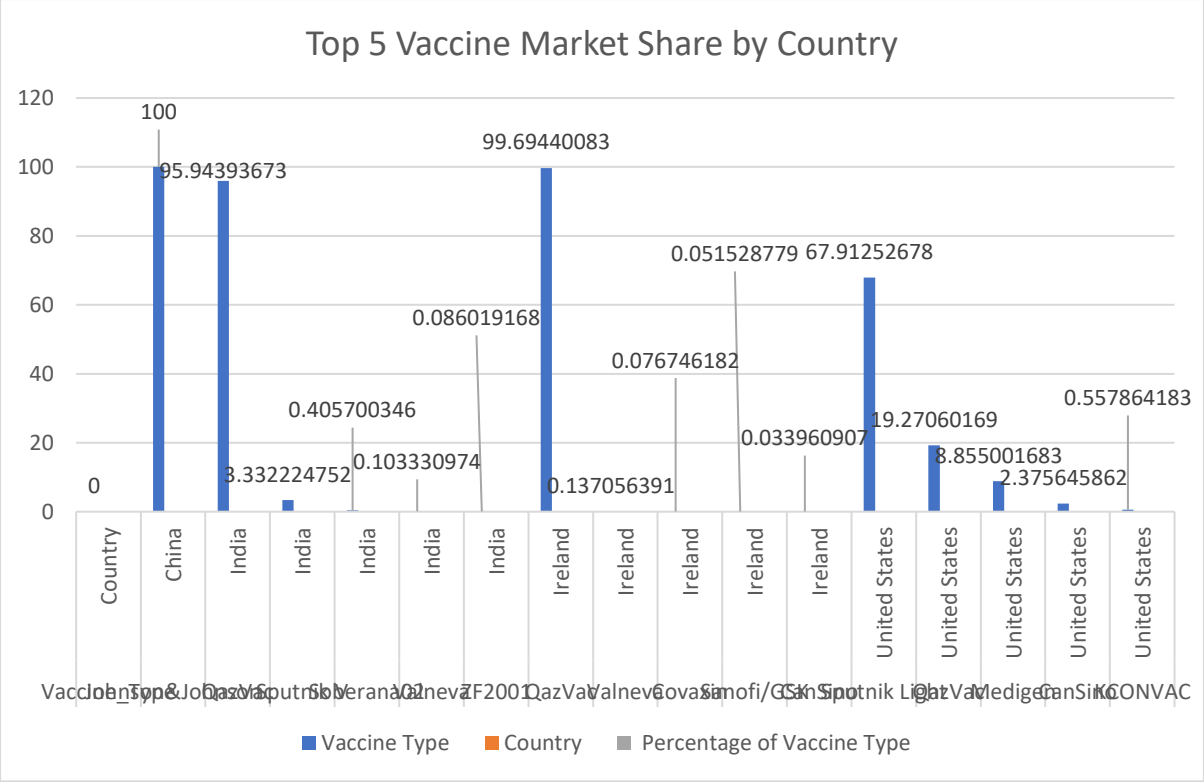
```
PercentageShare AS (
SELECT
Country_Name,
Vaccine_Type,
(Vaccine_Total * 100.0 / Country_Total) AS Percentage_Of_Vaccine
FROM
VaccineShare
),

RankedShare AS (
SELECT
Country_Name,
Vaccine_Type,
Percentage_Of_Vaccine,
ROW_NUMBER() OVER (PARTITION BY Country_Name ORDER BY Percentage_Of_Vaccine DESC) AS
Rank
FROM
PercentageShare
)

SELECT
Vaccine_Type,
Country_Name AS Country,
Percentage_Of_Vaccine AS "Percentage of Vaccine Type"
FROM
RankedShare
WHERE
Rank <= 5
ORDER BY
Country_Name, Percentage_Of_Vaccine DESC;
```



Vaccine_Type	Country	Percentage of Vaccine Type
Johnson&Johnson	China	100.0
QazVac	India	95.94393673129515
Sputnik V	India	3.332224751986735
Soberana02	India	0.4057003461415017
Valneva	India	0.10333097415581304
ZF2001	India	0.08601916766493031
QazVac	Ireland	99.69440083208417
Valneva	Ireland	0.13705639054628704
Covaxin	Ireland	0.07674618219918518
Sanofi/GSK	Ireland	0.05152877870191658 5
CanSino	Ireland	0.03396090668816059
Sputnik Light	United States	67.91252677775911



Task D.4.

```
WITH MonthlyTotals AS (
   SELECT
      c.Country_Name,
      strftime('%Y-%m', vr.Date) AS Month,
      ds.Source_Name || ' (' || ds.Source_URL || ')' AS Source_Name_URL,
      MAX(vr.Total_Vaccinations) AS Total_Administered_Vaccines
   FROM
      Vaccination_Record vr
   JOIN
      Country c ON vr.Country_ID = c.Country_ID
   JOIN
      Data_Source ds ON c.Country_ID = ds.Country_ID
   GROUP BY
      c.Country_Name, Month, ds.Source_Name, ds.Source_URL
)

SELECT
   Country_Name,
   Month,
   Source_Name_URL AS "Source Name (URL)",
   Total_Administered_Vaccines
FROM
   MonthlyTotals
ORDER BY
```

Total_Administered_Vaccines DESC;

Vaccination.db - Query

new_table_name Create Query

table name...

- Country
- Data_Source
- Manufacturer
- US_State_Vaccination
- Vaccination_AgeGroup
- Vaccination_Record
- Vaccine

Toggle helper tables

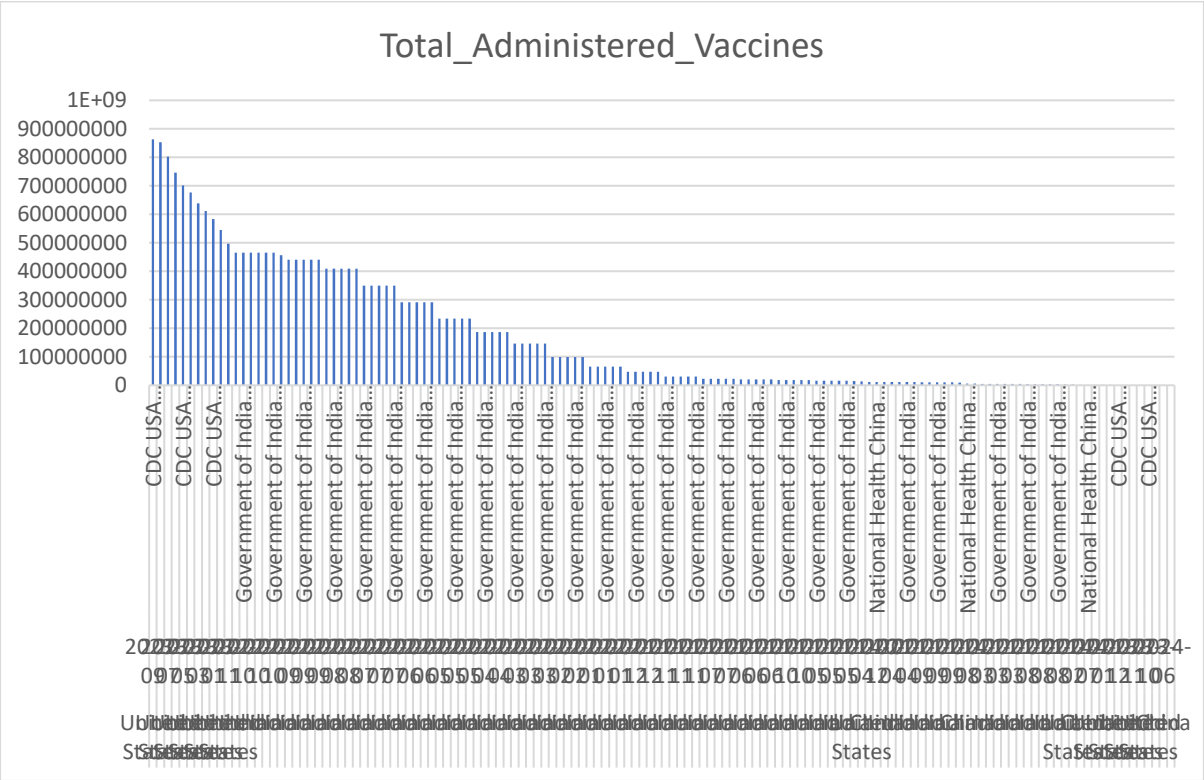
```
WITH MonthlyTotals AS (
  SELECT
    c.Country_Name,
    strftime('%Y-%m', vr.Date) AS Month,
    ds.Source_Name || ' (' || ds.Source_URL || ')' AS Source_Name_URL,
```

Use Shift + Up/Down to navigate recently-executed queries

Execute | Export JSON | Export CSV | SQL Help | Bookmarks | +

Results (unable to determine) Permalink

Country_Name	Month	Source Name (URL)	Total_Administered_Vaccines
United States	2023-09	CDC USA (https://data.cdc.gov/Vaccinations/COVID- ...	863237900
United States	2023-08	CDC USA (https://data.cdc.gov/Vaccinations/COVID- ...	852741029
United States	2023-07	CDC USA (https://data.cdc.gov/Vaccinations/COVID- ...	802334878
United States	2023-06	CDC USA (https://data.cdc.gov/Vaccinations/COVID- ...	746599967
United States	2023-05	CDC USA (https://data.cdc.gov/Vaccinations/COVID- ...	701902808
United States	2023-04	CDC USA (https://data.cdc.gov/Vaccinations/COVID- ...	677034638
United States	2023-03	CDC USA (https://data.cdc.gov/Vaccinations/COVID- ...	638402224
United States	2023-02	CDC USA (https://data.cdc.gov/Vaccinations/COVID- ...	611958849
United States	2023-01	CDC USA (https://data.cdc.gov/Vaccinations/COVID- ...	583013106
United States	2022-12	CDC USA (https://data.cdc.gov/Vaccinations/COVID- ...	545380681
United States	2022-11	CDC USA (https://data.cdc.gov/Vaccinations/COVID- ...	496803362
United States	2022-10	CDC USA (https://data.cdc.gov/Vaccinations/COVID- ...	465282194
India	2022-10	Government of India (http://mohfw.gov.in/pdf/Cumul ...	465161997
India	2022-10	Government of India (https://dashboard.cowin.gov.i ...	465161997

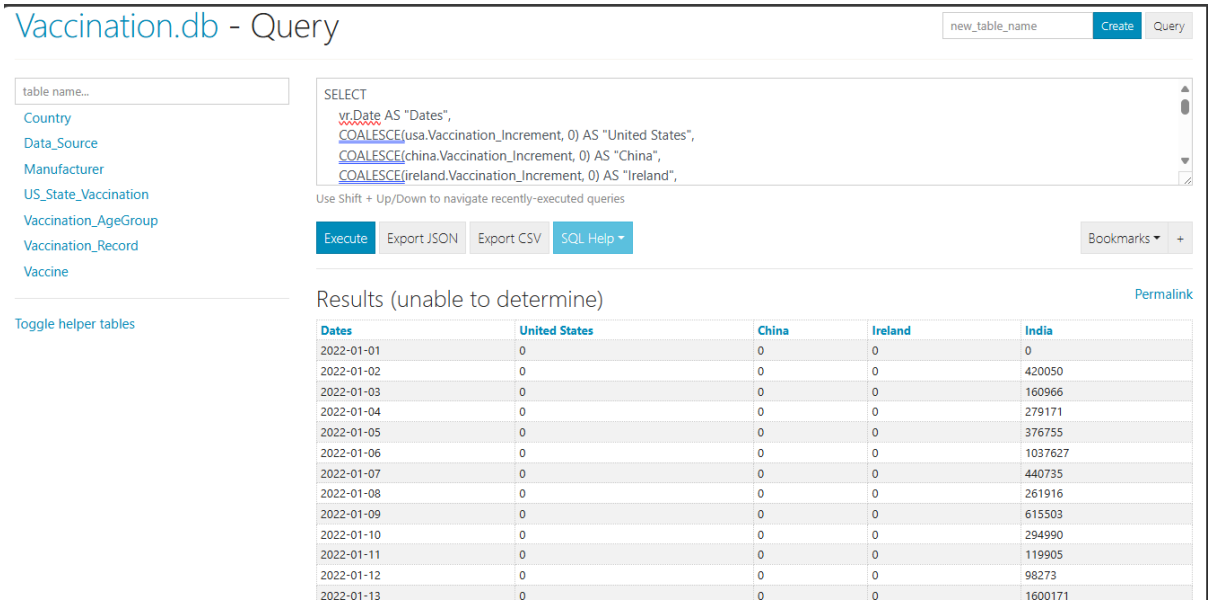
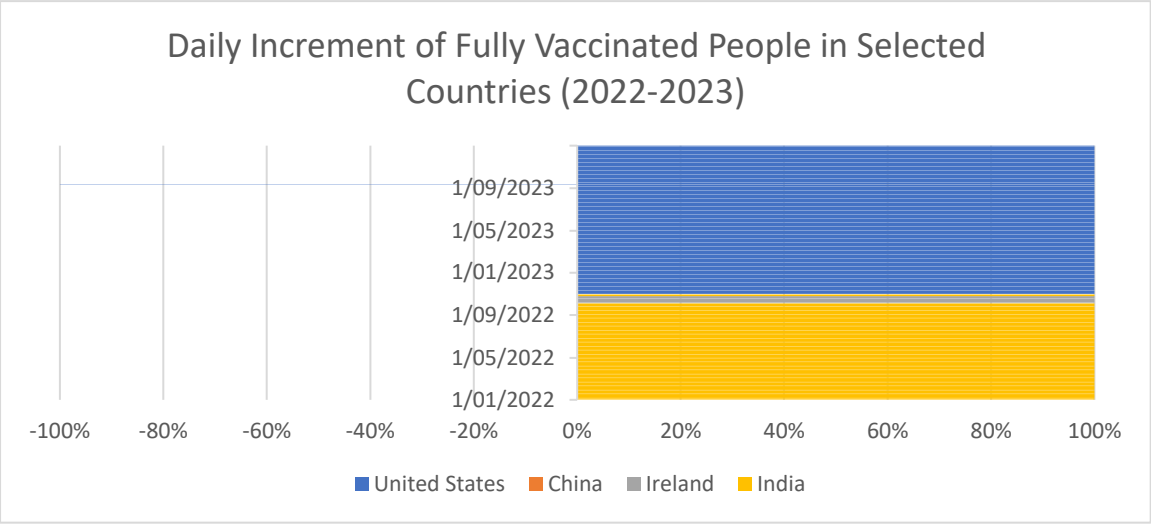


Task D.5.

```
SELECT
vr.Date AS "Dates",
COALESCE(usa.Vaccination_Increment, 0) AS "United States",
COALESCE(china.Vaccination_Increment, 0) AS "China",
COALESCE(ireland.Vaccination_Increment, 0) AS "Ireland",
COALESCE(india.Vaccination_Increment, 0) AS "India"
FROM
```

```
(SELECT DISTINCT Date FROM Vaccination_Record WHERE Date BETWEEN '2022-01-01' AND '2023-
12-31') vr
LEFT JOIN
(
SELECT
Date,
Total_Vaccinations - LAG(Total_Vaccinations, 1) OVER (ORDER BY Date) AS Vaccination_Increment
FROM
Vaccination_Record
WHERE
Country_ID = (SELECT Country_ID FROM Country WHERE Country_Name = 'United States')
AND
Date BETWEEN '2022-01-01' AND '2023-12-31'
) AS usa ON vr.Date = usa.Date
LEFT JOIN
(
SELECT
Date,
Total_Vaccinations - LAG(Total_Vaccinations, 1) OVER (ORDER BY Date) AS Vaccination_Increment
FROM
Vaccination_Record
WHERE
Country_ID = (SELECT Country_ID FROM Country WHERE Country_Name = 'China')
AND
Date BETWEEN '2022-01-01' AND '2023-12-31'
) AS china ON vr.Date = china.Date
LEFT JOIN
(
SELECT
Date,
Total_Vaccinations - LAG(Total_Vaccinations, 1) OVER (ORDER BY Date) AS Vaccination_Increment
FROM
Vaccination_Record
WHERE
Country_ID = (SELECT Country_ID FROM Country WHERE Country_Name = 'Ireland')
AND
Date BETWEEN '2022-01-01' AND '2023-12-31'
) AS ireland ON vr.Date = ireland.Date
LEFT JOIN
(
SELECT
Date,
Total_Vaccinations - LAG(Total_Vaccinations, 1) OVER (ORDER BY Date) AS Vaccination_Increment
FROM
Vaccination_Record
WHERE
Country_ID = (SELECT Country_ID FROM Country WHERE Country_Name = 'India')
AND
```

```
Date BETWEEN '2022-01-01' AND '2023-12-31'
) AS india ON vr.Date = india.Date
ORDER BY
vr.Date;
```



Dates	United States	China	Ireland	India
2022-01-01	0	0	0	0
2022-01-02	0	0	0	420050
2022-01-03	0	0	0	160966
2022-01-04	0	0	0	279171
2022-01-05	0	0	0	376755
2022-01-06	0	0	0	1037627
2022-01-07	0	0	0	440735
2022-01-08	0	0	0	261916
2022-01-09	0	0	0	615503
2022-01-10	0	0	0	294990
2022-01-11	0	0	0	119905
2022-01-12	0	0	0	98273
2022-01-13	0	0	0	1600171